

Jake Callcut

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TECHNICAL SKILLS

Languages and Technologies: Python, SQL, Java, Ada, Dart

Tools/Frameworks: Pandas, NumPy, scikit-learn, Git, Docker, APIs, JSON, OOP, Firebase

Other: Figma, RaspberryPi, Linux Environments, Autodesk Fusion 360

EDUCATION

Heriot-Watt University | MEng Software Engineering

09/2022 – 06/2027

- On Track for First Class Honours
- Twice awarded the Deputy Principles Award for academic excellence
- Relevant Modules: Rigorous Methods for Software Engineering, Hardware-Software Interfaces, Computer Network Security, Data Mining & Machine Learning, Data Structures & Algorithms

Torquay Boys' Grammar School

- **A Levels:** Computer Science (A), Mathematics (A), Physics (A)
- 09/2020 – 08/2022

RELEVANT EXPERIENCE

Squirrel | Software Engineer

06/2025 – Present

- Independently designed and developed the company's consumer mobile app and companion business platform, delivering end-to-end functionality from UI/UX to deployment, contributing to investor pitches and funding applications.

GI Healthcare Industries | Software Engineer

08/2024 – 11/2025

- Architected and implemented a multi-machine control system with a custom server-client design using Flutter, TypeScript, and Socket.IO, ensuring a high fault tolerance.
- Attended stakeholder product demonstrations, engaging with end-users to gather requirements, observe operational needs, and provide feedback to improve system reliability and usability.

Heriot-Watt University | Lab Assistant

09/2024 – 01/2025

- Assisted 25 students per week in debugging and refining their Java code, providing guidance on best practices

GI Healthcare Industries | UI/UX Design Intern

06/2024 – 08/2024

- Designed the front-end interface for a multi-machine control system, from low-fidelity wireframes to developer-ready handoff using Figma. Helping take the company's new product from TRL 3 to TRL 5.

PROJECTS & EXTRA-CURRICULAR

Social Sentiment Analysis for Sector Level Stock Market Prediction | Dissertation Project

- Designed and implemented a data pipeline to extract finance data, analyse social media posts for sentiment data using finBERT, and apply machine learning models to predict price movements across various market sectors.
- Evaluated model performance by analysing correlation and classification metrics to assess the predictive power of aggregated social sentiment.

Judging a Song by Its Cover | Machine Learning Project

- Implemented and compared multiple machine learning models and neural networks (kNN, Naive Bayes, MLP, and CNN)
- Developed algorithmic fine-tuning to optimise models for higher precision and F-Measure without over-fitting.

Lumina | Personal Project

- Engineered a smart home hub using Raspberry Pi and WebSockets, enabling reliable real-time communication between IoT devices and a central control system.
- Developed a Flutter-based web application to provide users with an intuitive interface for monitoring and controlling their smart home environment.

eTextiles Hackathon | The National Robotarium

- Worked with a multidisciplinary team to develop a JavaScript-based smart glove using microcontrollers to control devices.

ADDITIONAL INFORMATION

- Completed the Bloomberg Market Concepts course and applied skills using the Bloomberg terminal for market research.
- Completed online certification on Investment Analysis & Portfolio Management, edX.
- Fully DBS-checked mathematics tutor (GCSE, Highers, A Level).